

Supplementary material for: Predictive e-diagnostics for multi-state movement models

Aurélien Nicosia¹

¹Université Laval, Canada.

1 Simulation allocation

The main article reports the scenarios that directly support the central methodological narrative: calibration, targeted movement diagnostics, predictable localization and predictable diagnostic combination. This supplement records additional scenarios that are useful for sensitivity analysis, implementation warnings or methodological extensions.

Scenario	Placement	Reason
S1	Main article	Basic null calibration is required to support the e-process implementation.
S2	Supplement	Parameter estimation uses an oracle-state estimator in the current simulation, so it is best treated as a sensitivity check.
S3	Main article	The missing-state diagnostic is a core example of a full predictive HMM alternative.
S4	Main article	Angular misspecification is a central movement-specific diagnostic.
S5	Main article	Residual step-angle dependence is a central feature-level diagnostic.
S6	Main article	Duration misspecification is a key movement failure and illustrates blockwise observable features.
S7	Main article	Localization is a main theoretical and practical contribution.
S8	Main article	Predictable mixtures and switching are main combination tools.
S9	Supplement	The product example is an important warning, but it is not part of the recommended procedure.
S10	Supplement	Long-horizon straightness is an advanced blockwise extension beyond the main simulation narrative.
S11	Supplement	Individual cross-fitting is a validation-design extension.
S12	Supplement	Composite envelopes are conservative and optional for the first article.

Table 1 Allocation of simulation scenarios between the main article and supplementary material.

2 Simulation DGP details

To ensure the simulation results can be understood independently, we briefly outline the data-generating processes (DGPs) used in the main scenarios. Von Mises distributions in the simulations are parametrized through circular standard deviations (sd); the corresponding concentration parameters κ were obtained by numerical inversion of the circular variance formula $V = 1 - I_1(\kappa)/I_0(\kappa)$, where I_p denotes the modified Bessel function of order p . All parameters are specified on the natural scale.

- **S1 (Calibration):** A 2-state HMM with initial probabilities (0.5, 0.5) and transition matrix Γ with $\Gamma(1, 1) = 0.8$, $\Gamma(2, 2) = 0.7$. State 1 (encamped) has step lengths drawn from Gamma(shape = 2, rate = 0.5) and turning angles from von Mises($\mu = 0$, sd = 0.5). State 2 (exploratory) has step lengths Gamma(10, 2) and turning angles von Mises($\mu = 0$, sd = 2). The diagnostic q_t is an alternative 2-state HMM with slightly different parameters (e.g., transition and state-dependent parameters are perturbed relative to the null generator p_0).

- **S3 (Missing state)**: The true DGP is a 3-state HMM with an intermediate speed state, but the null model is restricted to $K = 2$. True transition matrix Γ : row 1 (0.91, 0.06, 0.03), row 2 (0.08, 0.87, 0.05), row 3 (0.08, 0.10, 0.82). Step shapes (2.0, 7.5, 4.0), rates (3.0, 2.0, 1.2), angle means (0, 0, 0.6), angle SDs (1.6, 0.35, 0.75). The diagnostic q_t uses the *true* 3-state DGP parameters (*oracle diagnostic*).
- **S4 (Angular misspecification)**: The true DGP has step shapes (2.0, 7.0), rates (3.0, 2.0), and angle SDs (1.8, 0.25). The null model is misspecified with angle SDs (0.85, 0.30) and state 2 centered at $\mu = 0.25$ rather than 0. The diagnostic q_t is a fixed, manually chosen diffuse-angle alternative (*fixed alternative*).
- **S5 (Copula dependence)**: The true DGP includes a Gaussian copula with correlation $\rho = 0.6$ between step lengths and turning angles within each state, violating the conditional independence assumption of standard HMMs. The copula diagnostic uses a fixed target correlation $\rho_q = 0.55$ (*fixed alternative*).
- **S6 (Duration)**: The true DGP is a hidden semi-Markov model (HSMM) where the dwell time in both states follows a shifted Poisson distribution (mean 18), rather than a geometric distribution. The duration diagnostic is a blockwise observable feature (switch counts within each block) and is *fixed* before seeing the data (*fixed alternative*).
- **S7 (Localized failure)**: Similar to S5 ($\rho = 0.8$ in state 2) but restricted to a time window (times 121 to 220). The diagnostic q_t uses a fixed predictable weight concentrated on the failure window (*fixed alternative*).
- **S8 (Mixture switching)**: The alternative has two successive defects: angular misspecification (times 1 to 149) followed by step-angle copula dependence (times 150 to 300). The mixture and switching diagnostics combine fixed alternatives specified before the run (*fixed alternatives*).

Future work could further characterize diagnostic sensitivity by exploring weaker alternatives, such as scenarios with highly overlapping state-dependent distributions, only slightly misspecified turning angles (e.g., small differences in concentration parameters), moderate copula dependence ($\rho \approx 0.1$), or near-geometric dwell time distributions.

3 Supplementary scenarios

3.1 S2: train/validation with estimated parameters

Scenario S2 compares known parameters, oracle-state estimated parameters evaluated under the true generator and oracle-state estimated parameters evaluated under the fitted generator. At $\alpha = 0.05$, the crossing rates are 0.057, 0.053 and 0.023, respectively. These results are compatible with the split-validation interpretation, but the estimator uses simulated latent states. For that reason, S2 is not used as a main article result.

Setting	Crossing rate	Monte Carlo SE	Mean final log e-value
Known parameters, true generator	0.057	0.013	-23.868
Oracle-state estimate, true generator	0.053	0.013	-23.885
Oracle-state estimate, fitted generator	0.023	0.009	-24.565

Table 2 S2 results at $\alpha = 0.05$.

3.2 S9: warning against naive parallel products

Scenario S9 demonstrates the consequence of multiplying diagnostics computed in parallel on the same validation observations. At $\alpha = 0.05$, a single valid e-process and the weighted average of two copies both cross at rate 0.030, while naive products of two and three copies cross at rates 0.117 and 0.215. This supports the recommendation to use predictable mixtures, switching or weighted averages rather than parallel products.

Method	Status	Crossing rate	Crossing rate minus α
Single e-process	Valid	0.030	-0.020
Weighted average of two copies	Valid	0.030	-0.020
Naive product of two copies	Not valid by default	0.117	0.067
Naive product of three copies	Not valid by default	0.215	0.165

Table 3 S9 results at $\alpha = 0.05$.

3.3 S10: blockwise long-horizon straightness

Scenario S10 uses block straightness, defined as net displacement divided by total path length, to detect a long-horizon failure generated by autocorrelated turning angles. At $\alpha = 0.05$, the crossing rate is 0.036 under the HMM null and 0.796 under the long-horizon alternative. This scenario supports the blockwise theory but is kept in the supplement because it is a more advanced extension.

Generator	Crossing rate	Monte Carlo SE	Mean final log e-value	Mean straightness
HMM null	0.036	0.012	-4.909	0.496
Autocorrelated-angle alternative	0.796	0.025	4.639	0.347

Table 4 S10 results at $\alpha = 0.05$.

3.4 S11: individual validation and cross-fitted averages

Scenario S11 studies validation by individuals. The weighted average of final individual e-values is conservative under the true and fitted null settings, with crossing rates 0.000 and 0.004 at $\alpha = 0.05$. The exploratory scan over any individual is not a globally valid e-value and crosses at rates 0.304 and 0.228 under the two null settings. Under a single failed individual, the cross-fitted average crosses at rate 0.996 and the individual scan crosses at rate 1.000.

Scenario	Method	Status	Crossing rate
True null	Cross-fitted final average	Valid final e-value	0.000
True null	Any individual scan	Exploratory	0.304
Fitted null	Cross-fitted final average	Valid final e-value	0.004
Fitted null	Any individual scan	Exploratory	0.228
Single failed individual	Cross-fitted final average	Valid final e-value	0.996
Single failed individual	Any individual scan	Exploratory	1.000

Table 5 S11 results at $\alpha = 0.05$.

3.5 S12: finite-family composite envelope

Scenario S12 studies a conservative envelope over a finite family of three HMM null generators. It is important to note that the practical use of the composite envelope (Proposition 17) is primarily intended for finite or compactly restricted families. Continuous envelopes of unbounded parameter families for densities such as Gamma or von Mises can become numerically unstable or computationally prohibitive. The center-only denominator is not uniform over the family and gives a crossing rate of 0.980 when the true generator is the diffuse-angle null. The composite envelope gives crossing rate 0.000 at $\alpha = 0.05$ for each generator in the finite family. The scenario illustrates uniform validity and its cost in conservatism.

Generator	Method	Status	Crossing rate
Center null	Oracle true null	Valid for known generator	0.020
Center null	Center-only denominator	Not uniform over family	0.020
Center null	Composite envelope	Uniform finite-family envelope	0.000
Persistent null	Oracle true null	Valid for known generator	0.027
Persistent null	Center-only denominator	Not uniform over family	0.000
Persistent null	Composite envelope	Uniform finite-family envelope	0.000
Diffuse-angle null	Oracle true null	Valid for known generator	0.043
Diffuse-angle null	Center-only denominator	Not uniform over family	0.980
Diffuse-angle null	Composite envelope	Uniform finite-family envelope	0.000

Table 6 S12 results at $\alpha = 0.05$.

4 Additional proofs

This section collects routine closure-property proofs that are used in the main article. The main text keeps the central density-ratio proof and the latent-state filtering argument, while the algebraic consequences are recorded here for completeness.

4.1 Optional stopping and anytime p-values

Proof of optional stopping and anytime-valid monitoring For the finite horizon $T < \infty$, Ville's maximal inequality applied to the nonnegative supermartingale $(E_{1:t})_{0 \leq t \leq T}$ gives

$$\mathbb{P}_0 \left(\max_{0 \leq t \leq T} E_{1:t} \geq \frac{1}{\alpha} \right) \leq \alpha.$$

The logarithmic version follows by monotonicity of the logarithm. For any stopping time τ , the stopped index $\tau \wedge T$ is bounded, so optional stopping for bounded nonnegative supermartingales gives

$$\mathbb{E}_0(E_{1:\tau \wedge T}) \leq \mathbb{E}_0(E_{1:0}) = 1.$$

Thus the stopped process is an e-value. □

Proof of the anytime p-value statement The event $\{\exists 0 \leq t \leq T : p_t^{\text{any}} \leq \alpha\}$ is identical to

$$\left\{ \sup_{0 \leq t \leq T} E_{1:t} \geq \frac{1}{\alpha} \right\}.$$

The result follows from the optional-stopping and anytime-valid monitoring bound. □

4.2 Diagnostic mixtures, switching and sequential workflows

Proof of predictable mixtures The mixture density

$$q_t^{\text{mix}}(y \mid \mathcal{F}_{t-1}) = \sum_{j=1}^J w_{j,t} q_{j,t}(y \mid \mathcal{F}_{t-1})$$

is predictable. Since each component is a subdensity and the weights are predictable with $\sum_j w_{j,t} \leq 1$,

$$\int q_t^{\text{mix}}(y \mid \mathcal{F}_{t-1}) \mu_t(dy) \leq \sum_{j=1}^J w_{j,t} \leq 1.$$

The predictive e-process theorem therefore applies. □

Proof of predictable switching Set $w_{j,t} = \mathbf{1}\{J_t = j\}$. Since J_t is $\mathcal{F}_{t-1}$ -measurable, these weights are predictable and sum to one. The result is the predictable-mixture result with a degenerate weight vector. □

Proof of sequential exploratory diagnostic workflows For each diagnostic stage j ,

$$\mathbb{E}_0(D_{1:j} \mid \mathcal{G}_{j-1}) = D_{1:j-1} \mathbb{E}_0(D_j \mid \mathcal{G}_{j-1}) \leq D_{1:j-1}.$$

Thus $(D_{1:m})_{0 \leq m \leq M}$ is a nonnegative supermartingale with respect to $(\mathcal{G}_m)$. Ville's inequality gives

$$\mathbb{P}_0 \left(\sup_{0 \leq m \leq M} D_{1:m} \geq \frac{1}{\alpha} \right) \leq \alpha.$$

□

Proof of the weighted average and parallel-diagnostic bound Linearity of expectation gives

$$\mathbb{E}_0 E^{\text{avg}} = \sum_{j=1}^J w_j \mathbb{E}_0 E^{(j)} \leq \sum_{j=1}^J w_j \leq 1.$$

Thus E^{avg} is an e-value. If $\max_j w_j E^{(j)} \geq 1/\alpha$, then $E^{\text{avg}} \geq 1/\alpha$, so Markov's inequality applied to E^{avg} gives the stated bound. The product warning follows because no conditional-independence or sequential-validity assumption has been imposed on the parallel factors. □

4.3 Localization and tempering

Proof of predictable weighted localization Since $E_t \geq 0$ and $0 \leq W_t \leq 1$,

$$E_t^{(W)} = (1 - W_t) + W_t E_t \geq 0.$$

Moreover,

$$\mathbb{E}_0(E_t^{(W)} \mid \mathcal{F}_{t-1}) = 1 + W_t \{\mathbb{E}_0(E_t \mid \mathcal{F}_{t-1}) - 1\} \leq 1.$$

The cumulative product is therefore an e-process by the same supermartingale argument used for predictive e-processes. $\square$

Proof of segment-specific localization Apply predictable weighted localization with $W_t = B_t \in \{0, 1\}$. $\square$

Proof of soft state-localized localization The filtered predictive probability $a_{t|t-1}(s)$ is $\mathcal{F}_{t-1}$ -measurable and lies in $[0, 1]$, so it is an admissible predictable localization weight. $\square$

Proof of power and linear tempering The linear case is predictable weighted localization with $W_t = \lambda_t$. For the power case, Jensen's inequality for the concave function $x \mapsto x^{\lambda_t}$ gives

$$\mathbb{E}_0(E_t^{\lambda_t} \mid \mathcal{F}_{t-1}) \leq \{\mathbb{E}_0(E_t \mid \mathcal{F}_{t-1})\}^{\lambda_t} \leq 1.$$

Both constructions therefore give conditional e-value increments. $\square$

4.4 Training, validation and composite nulls

Proof of split-sample conditional validity Conditional on $\mathcal{H}$, the fitted densities and diagnostic rules are fixed, and the validation filtration satisfies the assumptions of the predictive e-process theorem. The conditional anytime-valid bound follows from optional stopping. Taking expectations over $\mathcal{H}$ gives the unconditional bound. $\square$

Proof of the cross-fitted average For nonnegative weights satisfying $\sum_i w_i \leq 1$,

$$\mathbb{E}_0 E^{\text{cf}} = \sum_{i=1}^n w_i \mathbb{E}_0 E^{(i)} \leq \sum_{i=1}^n w_i \leq 1.$$

No independence among the e-values is required for this averaging argument. A product would require additional conditional-independence or sequential-validity assumptions. $\square$

Proof of the composite-null envelope For each fixed $\theta \in \Theta_0$, the envelope satisfies

$$p_\theta(y \mid \mathcal{F}_{t-1}) \leq p_t^{\text{env}}(y \mid \mathcal{F}_{t-1})$$

pointwise on the relevant support. Therefore,

$$\mathbb{E}_\theta(\bar{E}_t \mid \mathcal{F}_{t-1}) = \int \frac{q_t(y \mid \mathcal{F}_{t-1})}{p_t^{\text{env}}(y \mid \mathcal{F}_{t-1})} p_\theta(y \mid \mathcal{F}_{t-1}) \mu_t(dy) \leq \int q_t(y \mid \mathcal{F}_{t-1}) \mu_t(dy) \leq 1.$$

The cumulative product is therefore an e-process uniformly over the composite null family. $\square$

5 Reproducibility

The simulation scripts are stored in the project directory `simulations/`. The complete simulation suite is run by `simulations/run_all_simulations.R`. Generated outputs are written to:

- `results/simulation_tables/`;
- `results/simulation_figures/`.

These generated outputs are intentionally ignored by Git; they should be regenerated from the scripts when preparing a submission archive.